Research Note Short Communication

From Ambition to Action: Why International Climate Agreements Failed to Address the Systems Driving Climate Change & the Provisions They Should Have Included

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ABSTRACT

Although there is ample scientific evidence and the international community is aware of climate change as a global issue for many decades now, actions have been unable to restrict the temperature rise to 1.5 degrees centigrade. The reason behind the inability of the major environmental instruments such as Stockholm Declaration, Rio Declaration, Kyoto Protocol, Copenhagen Accord and Paris Agreement to bring about collective action has been identified in this essay. Each of these instruments has been analyzed separately as separate legal artefacts on the basis of seven criteria including legal status, bindingness, enforcement, incentive, implementation, equity and expected effect on emissions. It has been argued in this essay that the main problem with these instruments was their excessive dependence on voluntary national commitments and emission targets while neglecting the economic, technological and institutional infrastructure supporting fossil fuel dependency.

Four provisions from these instruments that would have made them more effective in tackling climate change have been identified in a policy analysis of international legal regime. With each provision comes the analysis of the possible political limitations and barriers of implementation that would have been faced along with any potential unintended consequences, including the possibility of increased enforcement leading to exit and shallow pledges rather than deep reductions. The lesson learned by this essay is that in order for action against climate change to be meaningful it is not enough for countries to set aggressive goals for themselves, but rather it is important to implement policies that change incentives, decrease structural dependence on high-emitting systems and provide accountability for unfulfilled pledges. Shifting the attention away from reduction of emissions once they have been generated and towards the structures responsible for their generation allows us to highlight the importance of implementation mechanisms within international environmental governance.

Keywords: Climate Governance | Policy | International Climate Agreements | Carbon Pricing

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1. INTRODUCTION

The physical basis of the problem has been understood for well over a century. Arrhenius (1896) calculated that changes in atmospheric carbon dioxide would alter surface temperatures and the evidence accumulated since has removed any serious scientific doubt about the direction of the effect. Yet, over a century later, the international response remains inadequate to the scale of the problem. International environmental conferences like the Stockholm Declaration, Rio declaration and Paris Agreement have taken crucial steps by recognizing climate change as a global issue and encouraging international cooperation. However, they consistently relied on voluntary commitments, vague action plans and weak enforcement mechanisms, preventing them from creating an effective framework capable of keeping global warming below 1.5°C above pre-industrial levels. The scale of the shortfall is measurable. The United Nations Environment Programme (2025) estimates that full implementation of current nationally determined contributions (NDCs) would produce 2.3–2.5°C of warming this century and that implementation of current policies alone would produce roughly 2.8°C. Global greenhouse gas emissions rose 2.3 per cent in 2024 to 57.7 gigatonnes of carbon dioxide equivalent (UNEP 2025). These projections have improved since Paris was adopted, when the equivalent figures stood at roughly 3–3.5°C, so the framework has had an effect; it has simply not had an effect of the required size.

This essay proposes four provisions that should have been included in the international agreements above to make global climate action more effective. It does so after first establishing what each agreement actually was – a declaration of principle, a binding protocol or a political accord are very different instruments, and their failures have different causes – and after considering the competing explanations that the existing literature offers for why climate cooperation has been so weak.

2. RESEARCH CONTEXT

We understand climate change to be a present and ongoing phenomenon that the nations of the world are forced to address and will have to manage for years to come. While many positive developments are occurring that may lessen the impact of climate change (such as the widespread adoption of alternative fuels and the growth of transit options that do not use fossil fuels), it is certainly true that many of those changes could have been made earlier, leading to an even lesser effect of climate change on planet Earth.

Using information on previous international environmental conferences, this essay explains what provisions the nations of the world could have put into international agreements that might have made climate change less impactful on the natural environment. What could they have agreed to that would have made a difference in keeping global warming below 1.5 degrees Celsius, the tipping point for long-term climate effects? What could they have done differently, or what did they not do that would have been important?

The analysis is explicitly counterfactual and carries a methodological cost which is stated here rather than buried. Counterfactual policy claims cannot be tested against observed outcomes. What they can be tested against is the historical record of what negotiators were willing to accept, what comparable provisions achieved in other treaty regimes and what the empirical literature says about the effect sizes of the policies concerned. Where a proposed provision would almost certainly have failed to secure agreement at the time, this essay says so.

Literature Review

The claim that international climate agreements have underperformed is not contested. Why they have underperformed is. The literature offers at least five explanations, and they carry different implications for what a better agreement would have looked like.

1. Collective action and free riding

The oldest explanation treats the atmosphere as a global commons and emissions abatement as a public good. Olson (1965) showed that large groups under-supply public goods because individual members capture only a fraction of the benefit they create and Hardin (1968) popularised the parallel argument for shared resources. Applied to climate, the implication is that every state has an incentive to let others abate. Barrett (2003) formalised this for treaty design, arguing that international agreements must be self-enforcing because no authority can compel sovereign states and that self-enforcing treaties therefore tend to be either broad and shallow or narrow and deep. Nordhaus (2015) proposed climate clubs with penalty tariffs on non-members as the remedy, on the grounds that participation must be made individually rational rather than merely virtuous. On this account, the missing ingredient in Stockholm, Rio and Paris is enforcement.

2. Distributive conflict and domestic politics

A more recent literature disputes the framing itself. Aklin and Mildenberger (2020) argue that climate politics is characterised less by international free riding than by distributive conflict inside states, between industries and constituencies that gain and lose from decarbonisation. On their reading, governments do not fail to act because they fear being exploited by other governments; they fail to act because domestic carbon-intensive interests can block action regardless of what others do. Colgan, Green and Hale (2021) develop a related argument around asset revaluation, holding that climate policy threatens the value of specific fossil-fuel assets and therefore mobilises concentrated, well-organised opposition. Putnam's (1988) two-level model supplies the mechanism: negotiators are constrained by what their domestic ratification set will bear, so international agreements converge on what the most constrained major party can accept. If this account is right, adding enforcement to a treaty does not solve the problem, because the binding constraint sits below the level at which the treaty operates.

3. Institutional fragmentation

Keohane and Victor (2011) characterise climate governance not as a regime but as a regime complex – a loosely coupled set of institutions with no clear hierarchy, spanning the UNFCCC, the multilateral development banks, trade bodies, plurilateral clubs and private standard-setters. They argue that fragmentation is the realistic equilibrium given divergent interests and that it delivers flexibility at the cost of coherence. Victor (2011) is blunter, arguing that the UN framework's insistence on universal consensus guaranteed that outcomes would track the least willing participant.

4. The depth-participation trade-off and selection effects

Downs, Rocke and Barsoom (1996) make an argument that is directly relevant to this essay's third provision. They observe that high compliance rates with international treaties are frequently used as evidence that international law works, when in fact states select into agreements whose requirements they were already going to meet. Deep cooperation, which requires states to do things they would not otherwise have done, is rare and requires enforcement precisely because it is deep.

5. Structural and technological lock-in

The fifth explanation and the one this essay builds on, is that emissions are produced by long-lived physical and institutional systems that policy has largely left untouched. Unruh (2000) describes "carbon lock-in" as the condition in which technological, infrastructural and institutional co-evolution around fossil fuels creates persistent barriers to alternatives, even where those alternatives are cost-competitive. Roads, grids, building stock, zoning codes, vehicle fleets and the industries built around them commit emissions decades in advance of any given negotiating round. This explanation does not displace the others. It complements the distributive account in particular, since the assets that constitute lock-in are the same assets whose owners resist revaluation. What it adds is a different target for policy: agreements that set output targets without addressing the systems generating the output are attempting to change a result while leaving its causes intact.

This essay's argument sits at the intersection of the lock-in and enforcement accounts. It holds that the instruments examined below failed on two fronts simultaneously – they lacked the authority to compel compliance with the targets they set, and they aimed at the wrong variable.

3. METHODOLOGY

The study's analytical framework is informed by existing scholarship on international environmental governance, environmental law and multilateral climate agreements. It draws on Handl's (2012) analysis of the Stockholm Declaration (1972) and Rio Declaration (1992) to examine the evolution of international environmental principles and the development of global environmental governance. The broader institutional, legal and policy dimensions of international environmental cooperation are informed by Axelrod, Downie and Vig's (2011) work on global environmental institutions and policy.The study also uses the work of Clémençon (2016), who analyses the Paris Agreement by examining the alternative perceptions about the Paris Agreement either as a breakthrough or as a failure and Dimitrov (2016), who analyses the negotiations behind the Paris Agreement and the role of political bargaining that led to its final terms. The treaties themselves and the decisions made in the Conference of the Parties are considered where there is the question of the nature of an obligation.

Each instrument is assessed against seven criteria, applied consistently:

- Legal status – whether the instrument is a treaty, a protocol, a declaration of principles, or a political statement with no formal adoption.
- Bindingness – whether mitigation obligations are obligations of result, obligations of conduct, or hortatory language.
- Enforcement capacity – whether anybody can determine non-compliance and attach consequences to that determination.
- Incentive structure – what makes compliance attractive: reputation, market access, finance, or the avoidance of penalty.
- Implementation mechanism – the reporting, review and verification machinery that turns commitment into observable behaviour.
- Equity implications – how the instrument distributes obligation between developed and developing states.

• Expected emissions impact – the plausible effect on global emissions, separated from the effect actually observed.

Each instrument is also placed in its historical setting, because the design choices that look like failures in retrospect were often the cost of securing international agreement at all.

The Instruments Assessed Individually

The Stockholm Declaration (1972)

Stockholm was the first UN conference which recognized the human environment as an object of international policy. The conference came up with twenty-six principles and an Action Plan, comprising 109 recommendations and directly resulted in the formation of the United Nations Environment Programme, which followed immediately after the conference.

The status of the Stockholm Declaration from a legal perspective was that of a non-binding instrument. There were no obligations whatsoever enshrined in any of its provisions, but Principle 21, which recognizes sovereign rights over natural resources and at the same time obliges not to inflict damage outside national jurisdictional borders, is considered today to be a customary international law provision (Handl 2012). The bindingness was zero and there was no enforcement mechanism; UNEP was established as a coordinating and catalyst organization without regulatory capacity. The incentive scheme was based on reputation only. The implementation scheme included recommendations directed at governments without requiring them to report back. With regard to equity, the Stockholm Conference was much more attentive than is often believed: Principles 9 through 12 stated that environmental problems in the developing world derive from underdevelopment, while Indira Gandhi’s address to the conference framed poverty itself as the greatest polluter, a framing that shaped the developing-country negotiating position for decades afterwards.

Its expected emissions impact was effectively zero, and this is the point at which historical context matters most. Climate change was not on the agenda at Stockholm in any meaningful sense. The first World Climate Conference was seven years away and the conference itself was conducted in a Cold War setting that saw the Soviet bloc boycott proceedings over the question of East German representation. Criticising Stockholm for failing to regulate greenhouse gases is anachronistic. Its relevant failure is structural rather than substantive: it established the template of the principled, unenforceable declaration and that template was reproduced for the next twenty years.

The Rio Declaration (1992)

Rio was the point at which climate became a treaty subject. The Declaration's twenty-seven principles were adopted alongside Agenda 21 and the UN Framework Convention on Climate Change was opened for signature at the same conference.

The Declaration's legal status was again non-binding, and its bindingness is best illustrated by its own drafting. Principle 16 provides that national authorities should endeavour to promote the internalisation of environmental costs, taking into account that the polluter should "in principle" bear the cost of pollution and adds the qualifier that this should occur without distorting international trade and investment. Three separate hedges appear in a single sentence. Principle 8, on reducing unsustainable patterns of production and consumption, is similarly framed as something states "should" do. Enforcement capacity rested with no one; the Commission on Sustainable Development, created in 1993, reviewed progress but could not

compel it. The incentive structure combined normative pressure with financial transfer through the restructured Global Environment Facility. The implementation mechanism under the accompanying UNFCCC introduced national communications and greenhouse gas inventories, which mattered, because it created the first shared factual basis for later negotiation.

On equity, Rio was the high-water mark. Principle 7 established common but differentiated responsibilities, acknowledging the different contributions of states to global environmental degradation and Principle 6 gave priority to the special situation of developing countries. The expected emissions impact was modest by design. The UNFCCC's Article 4.2 set an aim, not an obligation, for developed countries to return emissions to 1990 levels by 2000. Most did not and those that did largely owed the result to the post-1991 industrial contraction in Eastern Europe rather than to climate policy. Rio therefore demonstrates something specific: differentiated obligation and broad participation were achieved at the cost of any obligation with force.

The Kyoto Protocol (1997)

Kyoto is omitted from the original manuscript, which weakens the argument, because it is the one instrument that did most of what this essay recommends – and its history is the strongest available test of whether those recommendations would have worked.

Its legal status was that of a binding protocol, in force from February 2005. Its bindingness was genuine: Annex B parties accepted quantified emission limitation and reduction commitments averaging roughly five per cent below 1990 levels across 2008–2012. These were obligations of result. Its enforcement capacity was, by the standards of environmental treaties, considerable. The Compliance Committee established under decision 27/CMP.1 had a facilitative and an enforcement branch and the enforcement branch could declare non-compliance, require a compliance action plan, suspend a party's eligibility to use the flexible mechanisms and deduct the shortfall plus a further thirty per cent from the party's allowance in a subsequent commitment period. The incentive structure was market-based, through the Clean Development Mechanism, Joint Implementation and international emissions trading, which created the first substantial cross-border carbon markets. The implementation mechanism – national registries, annual inventories and expert review teams – remains the template the transparency framework uses today.

Its equity design, however, was the source of its collapse. Obligations fell on Annex I parties only. The United States Senate passed the Byrd-Hagel Resolution in 1997 by 95 votes to nil, resolving that the US should not sign any agreement imposing binding commitments on developed countries without comparable commitments for developing ones. The US never ratified. Canada, facing a shortfall it estimated would cost billions to purchase its way out of, withdrew in December 2011 under Article 27 and by doing so avoided the very penalty regime that was supposed to deter it. The expected emissions impact was modest at the outset and the observed impact was smaller still: Annex B parties collectively met their targets, but global emissions still rose by roughly sixty per cent between 1990 and 2012 and a meaningful share of the Annex B reduction reflected economic transition and the relocation of production rather than abatement.

Two conclusions follow and they pull in opposite directions. Kyoto shows that binding targets with attached consequences are constructible; it also shows that when they bind, states exit. Any argument for stronger enforcement has to answer that.

The Copenhagen Accord (2009)

The Accord is referenced in the original conclusion without being analysed and it is worth analysing because it is the hinge between two architectures. Its legal status is unusually weak: it was not adopted by the Conference of the Parties at all. A small group of leaders negotiated it outside the formal process and objection from a handful of parties meant the COP could only "take note" of it. Bindingness was nil. Enforcement capacity was nil. Its implementation mechanism was the innovation – countries inscribed self-determined pledges in appendices and accepted a framework of measurement, reporting and verification. Its incentive structure was financial, with a commitment to fast-start funding of thirty billion dollars for 2010–2012 and a goal of mobilising one hundred billion dollars annually by 2020, a target that was met only in 2022. On equity, its procedural exclusivity generated lasting resentment (Dimitrov 2010). Its expected emissions impact was consistent with roughly 3°C or more.

Copenhagen's significance is that the failure of top-down target-setting there produced the bottom-up design that Paris formalised. The pledge-and-review model was not chosen because anyone believed it was optimal; it was what remained after the alternative collapsed.

The Paris Agreement (2015)

Paris is a binding treaty containing largely non-binding mitigation obligations and the distinction is the key to assessing it. Its legal status is that of a treaty in force since November 2016. Its bindingness is procedural rather than substantive: Article 4.2 requires each party to prepare, communicate and maintain successive nationally determined contributions and to pursue domestic measures aimed at achieving them. The obligation is to have a target and to try, not to hit it. Article 4.3 requires successive contributions to represent a progression and the party's highest possible ambition, but the party itself judges what its highest possible ambition is.

Its enforcement capacity is deliberately minimal. Article 15 establishes a committee that is expert-based and facilitative and that must function in a manner that is transparent, non-adversarial and non-punitive. These words were negotiated, not accidental. The implementation mechanism is the strongest element of the agreement: the enhanced transparency framework under Article 13 and the five-yearly global stocktake under Article 14, the first of which concluded in 2023 with language on transitioning away from fossil fuels in energy systems. The incentive structure combines reputational pressure through transparency with market access under the Article 6 cooperative approaches and with climate finance under Article 9. On equity, Paris softened Rio's binary by qualifying common but differentiated responsibilities with the phrase "in the light of different national circumstances" and it has since added the loss and damage fund agreed in 2023, and a new collective finance goal agreed in 2024.

Its expected emissions impact should be stated fairly. Projections have fallen from around 3–3.5°Cat adoption to 2.3–2.5°C under full NDC implementation (UNEP 2025), so the architecture has produced movement. But UNEP also notes that recent improvements owe as much to methodological revision and to policy trends outside the treaty as to new pledges and that the United States notified withdrawal in January 2025 with effect a year later. Universality, the design's central virtue, has proved reversible. Chan (2016) identifies the underlying problem: because contributions are self-determined, each party applies its own conception of fairness and there is no procedure for reconciling incompatible conceptions or for establishing that the sum of them is adequate.

4. KEY FINDINGS

Provision One: Sustainable Urban Development

International agreements heavily focused on net national emissions targets but paid much less attention to how this goal should be achieved. They largely ignored the systems that have caused a technological lock in and dependence on fossil fuels, such as transport and city design. Unruh (2000) describes exactly this process: infrastructure, technology and the institutions built around them co-evolve into a configuration that resists substitution long after alternatives become viable. The way cities are designed determines whether people rely on private vehicles or low-carbon, eco-friendly alternatives for decades after they are built. Today, transport accounts for almost a fifth of global carbon dioxide emissions (Ritchie 2020). Rather than continuing to make driving a better experience by building highways and roads, international agreements should have encouraged governments to redesign urban areas in a way that minimized automobile reliance. For example, adopting a Transit-Oriented Development model (where cities are designed to be compact, walkable and centered around high-quality public transportation). The empirical basis for this is established: Cervero and Kockelman (1997) identified density, diversity and design as the built-environment characteristics associated with reduced vehicle travel and Newman and Kenworthy (1999) documented the relationship between urban density and per-capita transport energy use across cities.

Principle 8 of the Rio Declaration states that “States should reduce and eliminate unsustainable patterns of production and consumption” to achieve “sustainable development”. This principle needed to go a step further and required countries to adopt sustainable transport strategies, such as investing in public transport instead of roads and encouraging higher-density cities rather than spread out ones. By reducing transport emissions at their source rather than attempting to offset them afterwards, these measures would have led to permanent structural reductions in greenhouse gas emissions and made the transition away from fossil fuels much easier.

Three objections weaken this provision and the first is the most serious. Land use is governed at municipal and regional level in almost every state, while treaties bind national governments. A commitment to compact development would therefore have been a commitment that many signatories lacked the constitutional authority to deliver, which is a good part of why Principle 8 was drafted as vaguely as it was.

Second, the effect sizes are less dramatic than even their proponents typically suggest. Ewing & Cervero (2010) performed a meta-analysis and discovered that the elasticities of vehicle trips in response to changes in the built environment tend to be quite small, such that meaningful declines only happen in combination. Smart growth is effective, but incremental and incomplete; on its own, it would never have closed the gap to 1.5°C.

Third, there are distributional consequences. Investment in transit systems increases nearby land values, as Zuk et al. (2018) detail. It is not unreasonable to believe that an international duty to increase density, without measures ensuring affordable tenancy or increasing housing stock, would have generated a climate dividend at the expense of precisely those lower-income people who rely on public transportation most. This version of the idea would need to come paired with protections from unaffordable rents and prices, adding cost and negotiability.

Provision Two: International Carbon Pricing

Although improving urban planning to reduce the demands on fossil fuels is vital, there is an even greater need for economic incentive on behalf of governments to prevent the continued use of fossil fuels. It would be far easier to implement a universally standardized carbon tax regime at the outset rather than attempt to decarbonize the economies after several decades of carbon emissions, as it would be working with economic principles rather than fighting them. Setting a globally standardized minimum carbon tax rate would encourage consumers and firms to embrace greener energy sources many years ago, as it would make pollution cost more money. In addition, a globally standardized carbon tax rate would also encourage the early development of renewable energy technologies, allowing less greenhouse gas emissions in the process. This argument of adopting a global minimum pricing scheme as opposed to national programs is made by Cramton et al. (2017), who argue that negotiating the price is much easier compared to negotiating the quantities.

There have been several international agreements that have considered the implementation of a carbon tax but have not made any final decisions. In the Rio Declaration, the Polluter Pays Principle was put into effect in Principle 16 where the polluter should, in principle, bear the cost of pollution. But since the principle states a "should" rather than a "must", it has not been used much by most countries.

Three decades later, the gap can still be seen. According to the World Bank (2025), 28 percent of the total amount of greenhouse gas emissions in the world is being priced by a carbon price in eighty operating mechanisms, producing a revenue of more than one hundred billion dollars in 2024, with an average price of about nineteen dollars per tonne. The High-Level Commission on Carbon Prices concluded that consistency with the Paris temperature goal required prices of 40–80 dollars per tonne by 2020 and 50–100 dollars by 2030 (Stiglitz and Stern 2017). Coverage has grown substantially; the price level has not. Meanwhile the International Monetary Fund estimates global fossil fuel subsidies at around seven trillion dollars in 2022 once implicit environmental and health costs are counted (Black et al. 2023), which means much of the world prices carbon negatively.

However, taxation is the hardest competence to internationalize. No multilateral environmental body has ever been granted authority to set a domestic tax rate, and fiscal sovereignty is defended by states across every ideological position. A binding global floor price would have required governments to surrender an instrument they treat as constitutive of sovereignty and there is little in the negotiating record to suggest it was ever available.

Provision 3: Binding Compliance and Enforcement Measures

Economic incentives alone are not enough. International agreements should also contain stronger enforcement mechanisms, such as legal consequences for when countries ignore their commitments. While the flexibility and intentional open-endedness of the agreements have helped secure international cooperation, it also allowed governments (particularly those of developed countries who weren’t willing to take the lead on climate action) to implement virtually none of these policies with few significant consequences. For example, the Paris Agreement adopted a voluntary “contributions” (not commitments) model to ensure as many countries would participate as possible, but in doing so, ignored the fundamental question of how to ensure the final outcome would actually be fair or effective for the climate target. Because the pledges were voluntary and self-determined, each country would design their own standard

of what is “fair” to suit its own domestic political realities – increasing the chance that the collective result would fall drastically short of what is actually needed to stop warming (Chan, 2016)

Instead, countries that repeated failed to good on their commitments should have been required to make mandatory financial contributions to international climate adaptation funds and been suspended from participating in international carbon markets. Such measures would not simply punish countries after they failed but also encourage governments to prioritize reductions before targets were missed.

How penalties should be calibrated

By persistence rather than by size of shortfall alone, so that a party that misses once and corrects is treated differently from one that misses repeatedly.

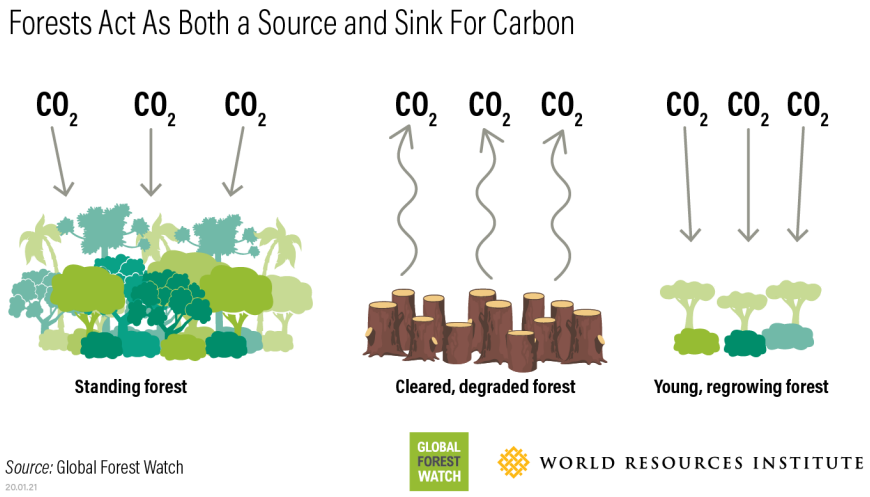
- First finding: a declaration of non-compliance and a compliance action plan submitted within ninety days, handled by the facilitative branch.
- Second consecutive finding: suspension of eligibility to transfer mitigation outcomes under Article 6.2 and to use Article 6.4 credits. This is self-executing, because it withdraws a privilege the treaty confers rather than imposing a new obligation – the distinction that let Kyoto's equivalent measure operate without a treaty amendment.
- Third or persistent finding: a mandatory contribution to the Adaptation Fund, set as excess tonnage multiplied by the volume-weighted average price across operating compliance carbon markets in the preceding year. That produces a figure which is transparent, externally determined and updated automatically. It should then be scaled by gross national income per capita and capped as a share of GNI and the shortfall carried into the next pledge period at a restoration rate above unity, following Kyoto's 1.3 precedent.
- Least developed countries and small island developing states exempt from the financial tiers and referred only to the facilitative branch, since the object is to deter avoidable non-compliance, not to penalise states lacking administrative capacity.

Provision 4: Conservation of Forests and Natural Carbon Sinks

Forests act as both a source and a sink for carbon – a double-edged sword, if you will. They can naturally remove about 30% of the carbon humans emit from burning fossil fuels annually (World Resources Institute 2025). But cutting them down releases carbon that has accumulated over decades or centuries.

As forests continue to be cleared for agriculture, infrastructure and urban expansion, this crucial natural carbon sink is becoming less effective. Analysis from WRI's Global Forest Watch and Land & Carbon Lab found that extreme fires in 2023 and 2024 cut forest carbon absorption to roughly a quarter of its typical level and that 2023 marked the weakest forest carbon sink in more than two decades (World Resources Institute 2025). The margin between forest emissions and forest removals is narrowing and the risk is a transition from sink to source.

Protecting forests, peatlands and wetlands is essential not only in keeping global warming in check, but in protecting thousands of endangered species and habitats.



International agreements should have included provisions that prohibited large-scale clearing of forests, required countries to achieve zero net deforestation through restoration and sustainable land management, promote conservation efforts and provide financial incentive funds for developing countries to reduce deforestation. If these provisions had been made earlier, thousands of acres of carbon-absorbing forest land could have been saved. Though the Paris Agreement encouraged countries to implement and support REDD+ (Reducing Emissions from Deforestation and Forest Degradation), it did not require them to do so and placed no penalties if they did not. Many climate policies focused primarily on emitting less in the future; whereas protecting forests would have helped absorb existing emissions as well. The pattern recurred in 2021, when more than 140 countries signed the Glasgow Leaders' Declaration on Forests and Land Use committing to halt and reverse forest loss by 2030. The declaration was political rather than binding and deforestation has not fallen at the rate the commitment implies.

That said, a treaty prohibition on clearing runs directly into the principle that Stockholm and Rio both affirm: the sovereign right of states to exploit their own natural resources. Forested states have consistently treated proposals of this kind as an attempt to convert their territory into a global asset without compensation, and the objection is not unreasonable given that deforestation in Europe and North America occurred largely before any such constraint existed.

There is also a human cost that conservation policy has repeatedly produced. Enforcement of protected areas has historically displaced forest-dwelling and indigenous communities whose tenure rights were unrecognized and a binding international prohibition on clearing, implemented by states with weak tenure systems, could reproduce that outcome at scale. A provision worth defending would have to tie protection obligations to recognition of community land rights rather than treating the two as separable.

5. DISCUSSION

The results point out that the primary deficiency of international climate agreements is not a failure to recognize the problem of climate change, but an absence of tools to alter the systems which generate the emissions. Although the Stockholm and Rio Declarations established the principles which could guide international cooperation in the environmental sphere and the Paris Agreement designed a model for climate policies decided at the national level, none of these models provided tools for implementing them

beyond the responsibility of the government. The four recommended provisions fill the gap by addressing the structural aspects of the generation of greenhouse gas emissions – urban design and transportation, economic incentives, compliance and protection of natural sinks of carbon dioxide.

The instrument-by-instrument assessment in Section 5 qualifies this conclusion in an important way. The failure was not uniform, and it is better described as a recurring trade-off than as a shared deficiency. Instruments that achieved broad participation and defensible equity provisions – Rio, Paris – did so by weakening obligation. The one instrument that imposed real obligations, Kyoto, lost its largest emitter at ratification and another party to withdrawal. Seen this way, the puzzle is not why negotiators failed to include enforcement, but why the conditions under which enforcement is sustainable have never obtained.

This is also where the competing explanations reviewed in Section 3 become consequential. If the binding constraint is international free riding, then enforcement is the answer and the provisions proposed here follow directly. If it is domestic distributive conflict, as Aklin and Mildenberger (2020) argue, then enforcement addresses a symptom: governments blocked by domestic coalitions cannot deliver and penalising them for not delivering produces exit rather than compliance. The Kyoto record is consistent with the second reading more than the first, since both the American and Canadian failures were driven by domestic politics rather than by fear of being exploited abroad. The honest position is that the provisions proposed here are necessary but insufficient and that their effectiveness depends on domestic political conditions this essay has not examined.

These proposals are counterfactual; thus, the results do not provide evidence about their adoption which would inevitably lead to avoiding 1.5°C warming. Stronger carbon prices and compliance would pose some difficulties related to distribution of responsibility and political challenges for developing countries and those whose economies depend on fossil fuels. Thus, the findings do not prove that binding rules were sufficient for solving the problem of climate change. In fact, it suggests that international agreements need a certain balance between flexibility and accountability. The analysis above suggests where that balance might lie strict compliance standards for procedural obligations, which states can control and which build the factual record on which everything else depends, combined with a due diligence standard and graduated, capacity-sensitive consequences for substantive ones.

6. CONCLUSION

The Stockholm Declaration, Rio Declaration, Copenhagen Accord and Paris Agreement – together with the Kyoto Protocol – each represented important milestones in international environmental cooperation. However, they ultimately relied too heavily on voluntary actions and where they did not, they lost participants. Had these agreements required sustainable urban planning, implemented a universal carbon pricing system, established penalties and legally protected natural sinks, the international community could have addressed both the causes and consequences of climate change much earlier. The difference between acknowledging climate change and addressing it lies in implementation – and that is where international agreements fall short. They didn't lack ambition, but rather the binding commitments, economic incentives and accountability needed to turn ambition into action.

Whether those commitments were ever politically available is a separate question and the record examined here gives reason for caution. What the record does establish is that the institutions capable of delivering them were never built and that the design problems standing in the way – who judges compliance, against what standard and at what cost – remain unsolved rather than merely unattempted.

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